

A COUNTEREXAMPLE TO THE CB-ISOMETRY YEADON-FACTOR CONJECTURE

AGENT LANE 14

STATUS

This packet gives a candidate counterexample to Conjecture 5.1 and to the “fundamental” Conjecture 5.2 of Arhancet [1]. For every $1 \leq p < \infty$, $p \neq 2$, it constructs finite von Neumann algebras M and N and a positive isometry

$$T : L^p(M) \longrightarrow L^p(N)$$

which is completely bounded, but whose Yeadon Jordan homomorphism J is not completely bounded. The same example also rules out the structural decomposition predicted in Conjecture 5.1.

SOURCE CONJECTURES

The source paper states the following conjecture in Section 5, page 7.

5 A conjecture on completely bounded isometries

We state the following conjecture for the structure of completely bounded isometries acting on noncommutative L^p -spaces associated with *finite* von Neumann algebras and $p \neq 2$.

Conjecture 5.1 *Suppose that M is a finite von Neumann algebra and that N is a semifinite von Neumann algebra. Suppose $1 \leq p < \infty$, $p \neq 2$. Let $T: L^p(M) \rightarrow L^p(N)$ be a (not necessarily surjective) isometry. Then T is completely bounded if and only if there exists a decomposition $M = M_1 \oplus M_2$ where M_1 and M_2 are von Neumann algebras and an integer $n \geq 1$ such that the restriction $T|_{L^p(M_1)}: L^p(M_1) \rightarrow L^p(N)$ is a complete isometry and such that the von Neumann algebra M_2 is n -minimal.*

The implication $\Leftarrow$ is true by the following reasoning. Note that using the decomposition of the normal Jordan $*$ -monomorphism of the isometry $T|_{L^p(M_2)}: L^p(M_2) \rightarrow L^p(N)$, we can write $T|_{L^p(M_2)} = T_1 + T_2$ where $T_1: L^p(M_2) \rightarrow L^p(N)$ and $T_2: L^p(M_2)^{\text{op}} \rightarrow L^p(N)$ are completely contractive. By (2.9), there exist some distinct integers $n_1, \dots, n_k \leq n$, some abelian von Neumann algebras $A_i \neq \{0\}$ and a $*$ -isomorphism

$$M = M_{n_1}(A_1) \oplus \dots \oplus M_{n_k}(A_k).$$

So we have an isometrical isomorphism

$$L^p(M_2) = S_{n_1}^p(L^p(A_1)) \oplus_p \dots \oplus_p S_{n_k}^p(L^p(A_k)).$$

It is folklore (and elementary) that the completely norm of the transpose map $t_n: S_n^p \rightarrow S_n^p$, $x \mapsto x^T$ satisfies $\|t_n\|_{\text{cb}, S_n^p \rightarrow S_n^p} = n^{|1-\frac{2}{p}|}$. Using this observation and [Jun, page 2], it is obvious that the identity map $\text{Id}_{L^p(M_2)}: L^p(M_2) \rightarrow L^p(M_2)^{\text{op}}$ is completely bounded. We infer that $T_2: L^p(M_2) \rightarrow L^p(N)$ is completely bounded. So the restriction $T|_{L^p(M_2)}: L^p(M_2) \rightarrow L^p(N)$ of T on $L^p(M_2)$ is completely bounded. Since $T|_{L^p(M_1)}: L^p(M_1) \rightarrow L^p(N)$ is completely contractive, we conclude that T is completely bounded.

For the reverse direction, it is quite easy to see that it suffices to prove the following.

Conjecture 5.2 *Suppose that M is a finite von Neumann algebra and that N is a semifinite von Neumann algebra. Suppose $1 \leq p < \infty$, $p \neq 2$. Let $T: L^p(M) \rightarrow L^p(N)$ be an isometry with a Yeadon factorization $T = wJ$. If T is completely bounded then the normal $*$ -Jordan homomorphism J is completely bounded.*

In text form, Conjecture 5.1 asks whether, for finite M , semifinite N and $1 \leq p < \infty$, $p \neq 2$, an isometry $T: L^p(M) \rightarrow L^p(N)$ is completely bounded if and only if there is a decomposition $M = M_1 \oplus M_2$ and an integer $n \geq 1$ such that $T|_{L^p(M_1)}$ is a complete isometry and M_2 is n -minimal. The source observes that the forward direction would follow from Conjecture 5.2: if $T = wJ$ is a Yeadon factorization and T is completely bounded, then the normal Jordan $*$ -homomorphism J should be completely bounded.

PROOF INTUITION

The transpose on S_m^p has completely bounded norm $m^{|1-2/p|}$, although it is an isometry at the scalar S_m^p level. The idea is to place both a homomorphic copy and a transposed copy of each matrix block in the range, but damp the transposed copy on the m th block by exactly $m^{-|1-2/p|}$. This damping is invisible to scalar isometry after a matching homomorphic weight is added in ℓ_p -sum, but it cancels the growth of the transpose cb norm at all matrix levels. Thus the isometry T is completely bounded.

The Yeadon Jordan part does not include the damping weights. It still contains the raw transpose on matrix blocks of arbitrarily large dimension, so it cannot be completely bounded. Since the same weighted map is not a complete isometry on any noncommutative matrix block,

the proposed complete-isometry/ n -minimal decomposition cannot be made: all large blocks would have to lie in the n -minimal side, but their dimensions are unbounded.

THE COUNTEREXAMPLE

Theorem 1. *Let $1 \leq p < \infty$ and $p \neq 2$. There exist a finite von Neumann algebra M , a finite hence semifinite von Neumann algebra N , and a positive isometry $T : L^p(M) \rightarrow L^p(N)$ such that:*

- (1) *T is completely bounded;*
- (2) *in the Yeadon factorization $T = wbJ$, the normal Jordan $*$ -homomorphism $J : M \rightarrow N$ is not completely bounded;*
- (3) *there is no decomposition $M = M_1 \oplus M_2$ with $T|_{L^p(M_1)}$ a complete isometry and M_2 n -minimal for some finite n .*

Consequently Conjectures 5.1 and 5.2 of [1] are false.

Proof. Put

$$\alpha = \left| 1 - \frac{2}{p} \right| > 0.$$

For $m \geq 2$, let tr_m be the normalized trace on $M_m(\mathbb{C})$ and let $\theta_m(x) = x^T$ be matrix transposition. Define

$$M = \prod_{m=2}^{\infty} M_m(\mathbb{C}), \quad N = \prod_{m=2}^{\infty} (M_m(\mathbb{C}) \oplus M_m(\mathbb{C})).$$

Equip them with the faithful normal finite traces

$$\tau_M((x_m)) = \sum_{m=2}^{\infty} 2^{-m} \text{tr}_m(x_m),$$

and

$$\tau_N((a_m, b_m)) = \sum_{m=2}^{\infty} 2^{-m} (\text{tr}_m(a_m) + \text{tr}_m(b_m)).$$

Both algebras are finite von Neumann algebras; in particular N is semifinite. The normalized trace only rescales each finite Schatten block, so it does not change the completely bounded norms of the block maps used below.

Let

$$d_m = m^{-\alpha}, \quad c_m = (1 - d_m^p)^{1/p} \quad (m \geq 2).$$

Thus $c_m, d_m > 0$ and $c_m^p + d_m^p = 1$. Define the normal Jordan $*$ -monomorphism

$$J((x_m)) = ((x_m, \theta_m(x_m)))_{m \geq 2} \quad (x = (x_m) \in M).$$

Let

$$b = ((c_m 1_m, d_m 1_m))_{m \geq 2} \in N_+, \quad w = J(1) = 1_N.$$

Since all c_m and d_m are strictly positive, $s(b) = 1_N = J(1) = w^*w$, and b commutes with $J(M)$. Now set

$$T(x) = wbJ(x) = ((c_m x_m, d_m x_m^T))_{m \geq 2}.$$

For $x = (x_m) \in L^p(M)$, transposition preserves the Schatten p -norm, and therefore

$$\begin{aligned} \|T(x)\|_{L^p(N)}^p &= \sum_{m=2}^{\infty} 2^{-m} (c_m^p \|x_m\|_{S_m^p}^p + d_m^p \|x_m^T\|_{S_m^p}^p) \\ &= \sum_{m=2}^{\infty} 2^{-m} (c_m^p + d_m^p) \|x_m\|_{S_m^p}^p = \|x\|_{L^p(M)}^p. \end{aligned}$$

Thus T is an isometry. The displayed formula also shows that $\tau_N(b^p J(y)) = \tau_M(y)$ for $y \in M_+$, so this is the Yeadon factorization in the convention recalled in [1, Section 2].

We next prove complete boundedness of T . The source paper records the standard formula

$$\|\theta_m : S_m^p \rightarrow S_m^p\|_{\text{cb}} = m^\alpha \quad [1, \text{p. } 7].$$

Fix a matrix level k and an element u in the k th amplification of the m th Schatten block. At this level the m th block of T satisfies

$$\|(c_m u, d_m \theta_m^{(k)} u)\|^p \leq (c_m^p + d_m^p m^{\alpha p}) \|u\|^p = (2 - m^{-\alpha p}) \|u\|^p \leq 2 \|u\|^p.$$

The matrix amplification of the direct product L^p -space is the corresponding weighted ℓ_p -sum of the amplified blocks. Summing the previous estimate over m gives

$$\|T\|_{\text{cb}, L^p(M) \rightarrow L^p(N)} \leq 2^{1/p}.$$

On the other hand J is not completely bounded. Indeed, the projection from N onto the second coordinate of the m th summand is completely contractive, and its composition with J restricted to the m th copy of M_m is exactly $\theta_m : M_m \rightarrow M_m$. The C^* -level cb norm of transposition on M_m is m . Hence $\|J\|_{\text{cb}} \geq m$ for every $m \geq 2$, which is impossible for a completely bounded map.

It remains to check that the structural conclusion of Conjecture 5.1 also fails. For each $m \geq 2$, the restriction

$$T_m : S_m^p \longrightarrow S_m^p \oplus_p S_m^p, \quad T_m(x) = (c_m x, d_m x^T),$$

is not a complete isometry. To see this, choose almost extremal vectors in matrix amplifications for the transpose norm. Letting the approximation error tend to zero gives

$$\|T_m\|_{\text{cb}} \geq (c_m^p + d_m^p m^{\alpha p})^{1/p} = (2 - m^{-\alpha p})^{1/p} > 1.$$

Since T_m is an isometry at scalar level, this strict amplification growth rules out complete isometry.

Any decomposition $M = M_1 \oplus M_2$ is given by a central projection, hence by a partition of the matrix factors M_m . If the m th factor, $m \geq 2$, lay in M_1 , then $T|_{L^p(M_1)}$ being a complete isometry would force T_m to be a complete isometry, a contradiction. Thus every factor M_m , $m \geq 2$, must lie in M_2 . But then M_2 contains full matrix algebras of arbitrarily large dimension, so it is not n -minimal for any finite n ; equivalently, it is not n -subhomogeneous for any finite n , using the equivalence quoted in [1, Section 2]. No decomposition as in Conjecture 5.1 exists. $\square$

VERIFICATION NOTES

The coefficient choice was checked by `code/check_block_estimates.py` for representative values of p and $2 \leq m \leq 80$. This is only a sanity check: the proof above uses the exact identities

$$c_m^p + d_m^p = 1, \quad d_m m^\alpha = 1, \quad c_m^p + d_m^p m^{\alpha p} = 2 - m^{-\alpha p} > 1.$$

The source crop was generated from the local PDF by `code/make_open_problem_crop.py`.

NOVELTY AND LIMITATIONS

The local lightweight indexes were searched:

```
registry_index.tsv
solutions/index.tsv
attempts/index.tsv
proof_gaps/index.tsv
```

The search terms were 2007.06999, the paper title, `completely bounded isometries`, `Yeadon factorization`, `normal Jordan`, and `n-minimal`. These searches found no exact duplicate and no existing answer packet for Conjectures 5.1–5.2.

External web searches on 2026-07-18 for the arXiv id and exact phrases around “If T is completely bounded then the normal Jordan homomorphism J is completely bounded”, “completely

bounded isometries Yeadon factorization”, and “normal Jordan completely bounded $L^p(M)$ isometry” found the source paper and related citations, but no later resolution of these conjectures. Novelty therefore remains candidate-level pending human or bibliographic review, but no known answer was found in the bounded search.

The main human-review point is the standard operator-space bookkeeping for matrix amplifications of direct sums of noncommutative L^p spaces. The counterexample otherwise uses only finite matrix blocks, positive scalar weights, and the transpose cb-norm formula explicitly recalled in the source.

REFERENCES

- [1] C. Arhancet, *A characterization of completely bounded normal Jordan $\ast$ -homomorphisms on von Neumann algebras*, arXiv:2007.06999, 2020.
- [2] F. J. Yeadon, *Isometries of non-commutative L^p -spaces*, Math. Proc. Cambridge Philos. Soc. 90 (1981), 41–50.